Vishal Gurjar

ep24btech11033@iith.ac.in | vishaldinesh844@gmail.com | +91 9256830303 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.41
Elite Global Sr. Sec. School, Kotputli	Class XII (RBSE)	2023	87.00%
Elite Global Sr. Sec. School, Kotputli	Class X (RBSE)	2021	97.00%

SKILLS

Programming languages: C, C++, Python, SQL.

Tools & Libraries: Git, Latex, Matplotlib, Numpy, Pandas.

Soft Skills: Time Management, Collaboration, Communication, Problem solving (LeetCode) .

RELEVANT COURSES

Computer Science: Introduction to Programming, Data Structures and Applications, Artificial Intelligence, Database Management Systems 1, Computational Physics.

Mathematics: Introduction to Probability, Introduction to Statistics, Elementary Linear Algebra, Differential Equations, Calculus, Mathematics for Physics.

PROJECTS

Student Grade Management System (Github Repo)

- Developed during the **Python and Machine Learning Workshop** organized by **StrategicERP** in association with the Electrical Engineering Department, IIT Hyderabad (Mentor: **Prof. Dr. Gajendranath Chaudhary**).
- Developed a **command-line Student Grade Management System** using **Python, MySQL, SQL, and OOPs** to manage a student's complete academic journey from school to multiple college degrees.
- Designed and integrated a MySQL relational database with CRUD operations, implemented SGPA, degree-wise CGPA, semester credit calculations, and academic report generation, and developed an interactive menu-driven interface with record/subject selection and field-wise update/delete workflows.

Eigenvalue Spectra of Financial Correlation Matrices (Github Repo)

(Under Prof. Sangkha Borah, Department of Physics, IIT Hyderabad)

- **Random Matrix Theory (RMT)**-based financial correlation analysis using **S&P 500** stock data, computation of **log returns**, **correlation matrix**, **eigenvalue decomposition**, and **Marchenko-Pastur distribution** comparison.
- **Data preprocessing**, **standardization**, **permutation testing**, **Wishart simulation**, **eigenvector stability**, **Inverse Participation Ratio (IPR)**, and statistical validation to separate meaningful correlations from noise.
- Identified **dominant market modes**, **outlier eigenvalues**, **top stock contributors**, and visualized the **eigenvalue spectrum** with statistical validation results.

Machine Learning Book Intern (Github Repo)

- Contributed to the *Machine Learning* educational book during a one-month (from May 29th to July 1st, 2025) internship under the supervision of **Prof. Ganesh Ramakrishnan (Department of CSE, IIT Bombay)**.
- Authored the complete **Prerequisites** section (approximately 100 pages) covering **NumPy, Pandas, and Matplotlib** with practical Python examples and beginner-friendly explanations.
- Prepared structured educational content including data visualization, array operations, matrix computations, data manipulation, and file I/O using Python.

Finite Difference Methods for Hyperbolic PDEs (Github Repo)

(Under Prof. Kirit Makwana, Department of Physics, IIT Hyderabad)

- Implemented and analyzed the **Lax-Friedrichs** and **Central Difference** schemes in Python to solve a coupled hyperbolic PDE initial value problem with zero-gradient boundary conditions.
- Compared the numerical stability and accuracy of both methods, demonstrating the instability of the Central Difference scheme and the stable behavior of the Lax-Friedrichs method through simulations and animations.

POSITIONS OF RESPONSIBILITY

Core Member, Torque Club, IIT Hyderabad (2025-26): Contributed to the planning, coordination, and execution of club activities and events while collaborating with the core team.

Served as **Student Coordinator**, facilitating communication between **Prof. Pawan Goyal**, TAs, and IIT Hyderabad students for the course Algorithms for Data Analysis and Text Processing at **IIT Kharagpur**.

EXTRACURRICULAR

- **NCC Cadet (B Certificate)** - Conduct theory and practical training sessions for junior cadets at IIT Hyderabad.
- **Spiritual Reading** - Regularly study the Bhagavad Gita and other spiritual texts.